

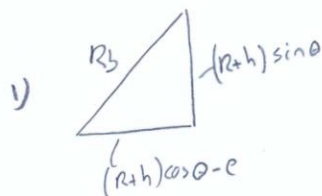
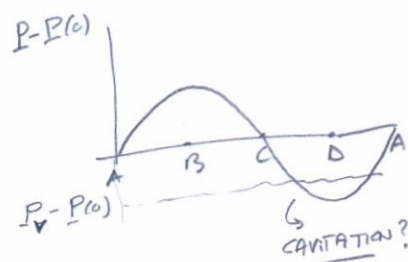
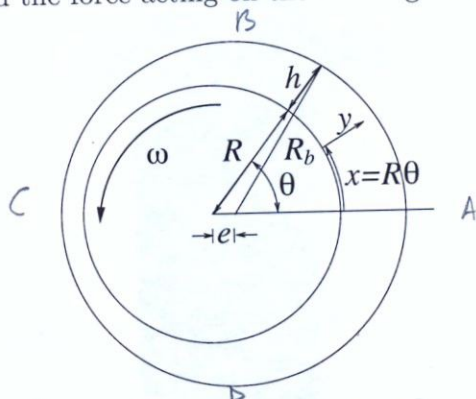
VF 12

The cylindrical bearing shown in the figure includes an external fixed casing of radius R_b and a shaft rotating with angular velocity ω of radius R such that $R_b - R \ll R$. The centers are shifted a small distance e , called excentricity, so that in between both cylinders there exists a thin circular gap of thickness $h \sim e \sim R_b - R \ll R$ full of a liquid of density ρ and viscosity μ .

1. Show by simple trigonometric reasoning that the gap thickness can be expressed as

$$h(\theta) = R_b - R + e \cos \theta.$$

2. Obtain the azimuthal velocity distribution v_x in terms of the distance y to the fixed casing and the unknown value of the induced pressure gradient $P_l = -R^{-1} \partial P / \partial \theta$.
3. Determine the circulating volume flux $Q = \int_0^h v_x dy$ as well as the pressure distribution $P(\theta) - P(0)$, where $P(0)$ is the unknown value of the pressure at $\theta = 0$, which depends on the pressurization conditions.
4. Calculate the torque and the force acting on the rotating shaft.



$$R_b^2 = (R_b - R + R)^2 = (R + h)^2 - 2e(R + h) \cos \theta + e^2$$

$$(R_b - R)^2 + 2(R_b - R)R = R^2 + 2Rh + h^2 - 2Re \cos \theta - 2eh \cos \theta$$

$$e \sim h \sim R_b - R \ll R$$

$$h = R_b - R + e \cos \theta = (R_b - R) [1 + \epsilon \cos \theta]$$

$$v_x = \omega R \frac{(h-y)}{h} = \frac{y(h-y)}{2\mu} \frac{1}{R} \frac{\partial P}{\partial \theta} \rightarrow Q = \int_0^h v_x dy = \frac{\omega R h}{2} - \frac{h^3}{12\mu} \frac{1}{R} \frac{\partial P}{\partial \theta} \Rightarrow \frac{\partial P}{\partial \theta} = \frac{12\mu R}{h^3} \left(\frac{\omega R h}{2} - Q \right)$$

$$\text{INTEGRATING ONCE} \quad P(\theta) - P(0) = 12\mu R \left(\frac{\omega R}{2} \int_0^\theta \frac{d\theta}{h^2} - Q \int_0^\theta \frac{d\theta}{h^3} \right)$$

$$\text{SINCE } P(2\pi) = P(0) \Rightarrow Q = \frac{\omega R}{2} \frac{\int_0^{2\pi} \frac{d\theta}{h^2}}{\int_0^{2\pi} \frac{d\theta}{h^3}} = \frac{\omega R (R_b - R)}{2} \frac{\int_0^{2\pi} \frac{d\theta}{(1 - \epsilon \cos \theta)^2}}{\int_0^{2\pi} \frac{d\theta}{(1 - \epsilon \cos \theta)^3}} = \omega R (R_b - R) \frac{1 - \epsilon^2}{2 + \epsilon^2}$$

$$P(\theta) - P(0) = \frac{6\mu \omega R^2}{(R_b - R)^2} \frac{\epsilon (2 + \epsilon \cos \theta) \sin \theta}{(2 + \epsilon^2)(1 + \epsilon \cos \theta)^2}$$

$$P(\theta) = -P(-\theta) \Rightarrow \text{VERTICAL FORCE}$$

$$P_{\min} \text{ AT } \theta = \frac{3\pi}{2} \Rightarrow \text{CAVITATION!}$$

FORCE ON THE SHAFT

$$F_z = - \int_0^{2\pi} (P - P(0)) n_z R d\theta = -R \int_0^{2\pi} (P - P(0)) \sin \theta d\theta = - \frac{12\mu \omega R^3 \pi}{(R_b - R)^2} \frac{\epsilon}{(1 - \epsilon^2)^{1/2} (2 + \epsilon^2)}$$

$$\text{FORCE ON THE BEARING} + \frac{12\mu \omega R^3 \pi}{(R_b - R)^2} \frac{\epsilon}{(1 - \epsilon^2)^{1/2} (2 + \epsilon^2)}$$

$$\text{TORQUE } \tau_p = \int_0^{2\pi} \tau_p R d\theta = - \frac{4\pi \mu \omega R^3}{R_b - R} \frac{(1 + 2\epsilon^2)}{(1 - \epsilon^2)^{1/2} (2 + \epsilon^2)}$$

BEARINGS CAN IN PRINCIPLE WITHSTAND AN INFINITE FORCE BECAUSE AS $F_z \rightarrow \infty$, $\epsilon \rightarrow 1$.

HOWEVER, FAILURE APPEARS WHEN THE MINIMUM PRESSURE REACHES P_v AND THE LIQUID CAVITATES.